TOPIC 01 · EXCEL FUNDAMENTALS WITH COPILOT
LAB 3 THEME · PIVOTTABLES, PIVOTCHARTS & VISUALS

# Lab 3 — Build PivotTables, PivotCharts and Recruitment Charts with Copilot

| | |
|---|---|
| **Maps to** | LO1: use Copilot to build PivotTables, PivotCharts and trend, comparison, funnel and combo charts from prompts, refine them with follow-up prompts and tie each chart to a checked total |
| **Duration** | 45 minutes |
| **Tools** | Excel · Copilot (Allow editing) · PivotTables, PivotCharts · line, bar, funnel and combo charts · BrightPath_Recruitment_Metrics.xlsx |
| **Mock data** | data/BrightPath_Recruitment_Metrics.xlsx |
| **Course** | Microsoft 365 Copilot for Excel (C34) · v3.2 |
| **Prompts** | https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-3 |

## Goal

Produce four decision-ready recruitment charts - hiring trend, hires by sector, the H1 2026 funnel and source effectiveness - each backed by a reconciled summary.

## What You Will Do

BrightPath's leadership wants to see how hiring is moving and which sectors and sourcing channels perform best. You will ask Copilot to build PivotTables and PivotCharts from 18 months of funnel metrics and a sourcing summary, choose each chart from the question it answers, refine titles, labels, sort order and axes with follow-up prompts, and reconcile every chart to an independent SUMIFS check.

## What You Will Build

BrightPath_Recruitment_Metrics.xlsx with PivotTables and four refined charts on the Charts sheet: a monthly hires line PivotChart, a sorted 2026 hires-by-sector bar PivotChart, a 2026 recruitment funnel and a hire-rate vs cost-per-hire combo chart

## Prerequisites

- Labs 1-2 completed: you can open a lab workbook, switch Copilot modes and verify with formulas.
- This lab folder's data/BrightPath_Recruitment_Metrics.xlsx.

**Data note**

Use only the supplied fictional BrightPath data or information you are authorised to process. Do not paste credentials, confidential business material or real employees' personal data into Copilot, rules or skills.

## Steps

**Step 1.** Copy BrightPath_Recruitment_Metrics.xlsx from this lab's data folder into a work-c34 folder on your computer and open it in Excel for Microsoft 365 (desktop). Work on this local copy - no OneDrive upload is needed. Check that Formulas > Calculation Options is Automatic, then open Copilot from the Copilot button on the Home tab or the floating Copilot icon at the bottom right. Read the About_This_Data sheet before you prompt. The Charts sheet is empty: you will create each chart there with a prompt, in a 2 x 2 grid. The data is on Monthly_Metrics and Source_Summary. Each chart answers one of the four questions below.

**PROMPT / CHECK — copy and adapt**

```
Tables : tblMonthly (108 rows = 18 months x 6 sectors) | tblSources (7 rows)

Q1 Trend : How have total hires moved month by month since Jan 2025?
Q2 Comparison : Which sector produced the most hires in Jan-Jun 2026?
Q3 Funnel : Where do candidates drop out between application and hire in 2026?
Q4 Efficiency : Which sources convert best, and at what cost per hire?
```

**Step 2.** Choose Allow editing. Ask Copilot to create the trend PivotTable and a line PivotChart on the Charts sheet. Then ask Copilot to check the Pivot grand total against =SUM(tblMonthly[Hires]) and name the peak month.

**PROMPT / CHECK — copy and adapt**

```
Create a PivotTable from tblMonthly on a new sheet named Pivot_Trend with Month_Label in rows
and Sum of Hires as values. Then create a line PivotChart of total hires by month on the
Charts sheet, placed over A1:H18. Format it cleanly: white background, no border or effects,
one blue line with markers, data labels above the points, light grey gridlines, axis titles
Month and Hires.
Title: "Monthly hires, Jan 2025 - Jun 2026".

CHECK (Allow editing):
In an empty cell on Pivot_Trend, add this formula. Tell me whether it matches the PivotTable
grand total, and which month had the most hires:
=SUM(tblMonthly[Hires])
```

**Step 3.** Ask Copilot to create the 2026 sector comparison chart on the Charts sheet. Then refine it with a follow-up prompt. Ask Copilot to add the SUMIFS check for Technology's 2026 hires.

**PROMPT / CHECK — copy and adapt**

```
Create a PivotTable on a new sheet Pivot_Sector from tblMonthly with Sector in rows and Sum of
Hires, filtered to months in 2026 only. Then create a bar PivotChart from it on the Charts
sheet, placed over J1:Q18, in the same clean style as the first chart.

FOLLOW-UP: Sort the bars from highest to lowest and change the title to state the leading
sector.

CHECK (Allow editing):
In an empty cell on Pivot_Sector, add this formula and tell me if it matches Technology's bar:
=SUMIFS(tblMonthly[Hires],tblMonthly[Sector],"Technology",tblMonthly[Month],">="&DATE(2026,1,1))
```

**Step 4.** Ask Copilot to build a 2026 funnel summary with formulas on its own sheet, then create a funnel chart on the Charts sheet. Ask Copilot to show the formula in each value cell to confirm the summary uses SUMIFS, not typed numbers.

**PROMPT / CHECK — copy and adapt**

```
On a new sheet named Funnel_2026, build a small table of 2026 totals for Applications,
Screened, Interviews, Offers and Hires from tblMonthly using SUMIFS on Month >= 1 Jan 2026.
Add a column showing each stage as a % of Applications. Then create a funnel chart from it
with data labels on the Charts sheet, placed over A20:H37, in the same clean style as the
first chart.

FOLLOW-UP: Show the formula in each value cell of the funnel table.
```

**Step 5.** Ask Copilot to add rate columns to tblSources and create a combo chart with two axes on the Charts sheet. Then ask it for the best hire rate and the lowest cost per hire.

**PROMPT / CHECK — copy and adapt**

```
In tblSources add Hire_Rate = Hires / Applications (percentage, 1 decimal) and Cost_per_Hire =
Sourcing_Spend / Hires (S$, no decimals). Create a combo chart on the Charts sheet:
Hire_Rate as clustered columns sorted highest first, Cost_per_Hire as a line on a secondary
axis.
Label both axes with units. Place it over J20:Q37 in the same clean style as the first chart.

FOLLOW-UP: Which source has the highest Hire_Rate, and which has the lowest Cost_per_Hire?
Give both values.
```

**Step 6.** Refine the four charts on the Charts sheet with short follow-up prompts, one change at a time. Add alt text to each chart.

**PROMPT / CHECK — copy and adapt**

```
Make every chart title state the finding in one sentence.
Use the same blue colour for all hires series and format numbers with thousands separators.
Make the four charts on the Charts sheet the same size and line them up in a 2 x 2 grid.
Add alt text to each chart describing what it shows.
```

**Step 7.** Ask Copilot (Chat only) to review the charts against the four questions, then ask it to log your prompts and checks on Prompt_Log.

**PROMPT / CHECK — copy and adapt**

```
CHAT ONLY:
Review the four charts on the Charts sheet. For each, tell me which question it answers (Q1
Trend, Q2 Comparison, Q3 Funnel, Q4 Efficiency), whether the chart type suits that question,
and one improvement. Do not change anything.

LOG (Allow editing):
Prompt_Log already lists every prompt in this lab. For each one I ran, fill in What Copilot
did and How I verified, and set Decision to Keep. I will tell you any row to change to Revise
or Undo.
```

## Sample Prompts

Use these prompts in the Copilot pane (or adapt them). Name the table and columns, state the output, and always verify the result. The same prompts are in sample-prompts.txt in this lab folder.

**Copy-ready prompts online:**
https://tertiarycourses.github.io/C34-Microsoft-365-Copilot-for-Excel/prompts/#lab-3

| # | Purpose | Prompt |
|---|---|---|
| 1 | PivotTable | Create a PivotTable from tblMonthly with Month_Label in rows, Sector in columns and Sum of Hires as values. |
| 2 | PivotChart | Create a line PivotChart from the Pivot_Trend PivotTable showing total hires by month. Show data labels and gridlines. |
| 3 | Bar chart | Create a bar chart of 2026 hires by Sector, sorted from highest to lowest. |
| 4 | Funnel | Create a funnel chart of 2026 Applications, Screened, Interviews, Offers and Hires with data labels. |
| 5 | Combo chart | Create a combo chart of Hire_Rate as columns and Cost_per_Hire as a line on a secondary axis. |
| 6 | Refine | Change the chart title to: Technology led hiring in H1 2026. |
| 7 | Refine | Sort the bars from highest to lowest and add data labels formatted with thousands separators. |
| 8 | Accessibility | Add alt text to this chart that describes the trend it shows. |
| 9 | Review | Is a line chart the right choice for this data? Suggest a better chart if not, and explain why. |

## Test It

| Check | Expected result |
|---|---|
| Total hires (Pivot_Trend) | 1,065 = SUM(tblMonthly[Hires]) |
| Peak month | 2025-11 with 68 hires |
| 2026 hires by sector | Technology first (84); Pivot total 361 |
| Funnel | 6,616 > 3,229 > 1,385 > 495 > 361 |
| Best source | Referral: highest hire rate (8.7%); MyCareersFuture: lowest cost per hire (S$126) |
| Charts sheet | 2 PivotCharts + funnel + combo in a 2 x 2 grid (A1:Q37), clean style, finding-style titles, labelled axes and alt text |

## Checkpoint

Keep the workbook in work-c34. Topic 2 starts from the client orders workbook in the Lab 4 folder.

## Troubleshooting

| If this happens | Fix |
|---|---|
| The 2026 filter includes 2025 months. | Ask Copilot to filter on the Month date field (>= 1 Jan 2026), not on Month_Label text, or add a Year field. |
| Copilot inserts a picture of a chart instead of a chart. | That came from a Chat answer and is static. Switch to Allow editing and ask it to create a chart linked to the data. |
| The funnel chart type is not offered. | Funnel charts need Excel 2019 or Microsoft 365. Ask for a sorted horizontal bar chart of the stages instead. |
| A chart lands on the PivotTable sheet, not on the Charts sheet. | Ask: 'Move this chart to the Charts sheet over A1:H18' (use that chart's range), or select the chart and use Move Chart > Object in: Charts. |
| The chart has a glow or gradient border, or the data labels overlap. | Ask: 'Remove the chart border and effects, use a white background and put the data labels above the points.' |
| Both series in the combo chart share one axis and the line is flat. | Ask: 'Put Cost_per_Hire on a secondary axis and format it as S$'. |

## Challenge

Ask Copilot for a 100% stacked column chart showing each sector's share of hires by quarter for 2025-2026, then decide whether it communicates better or worse than the bar chart, and why. Then try creating a spreadsheet from scratch: in a new blank workbook ask Copilot 'Create a spreadsheet I can use to plan and track BrightPath's July hiring campaign, with a Gantt chart visual.' Change one start date and check that the Gantt bar moves.

## Reflection

Which of your four charts would most change a BrightPath leadership decision, and what check makes you trust it?